



Cloud Bites

AWS-Based Restaurant Ordering System

Major Project

Team Members:

Arpit Gupta & Aditya Shukla

Faculty Guide

Anmol Singh

Problem Statement

The Challenges

- Manual order management & paper records
- Slow processing & high human error risk
- Difficult menu & inventory management
- Lack of centralized customer data

The Solution: Cloud Bites

A modern, AWS-powered cloud infrastructure that automates ordering, streamlines admin tasks, and provides a scalable, secure database architecture.

Project Overview

 Deployed on AWS

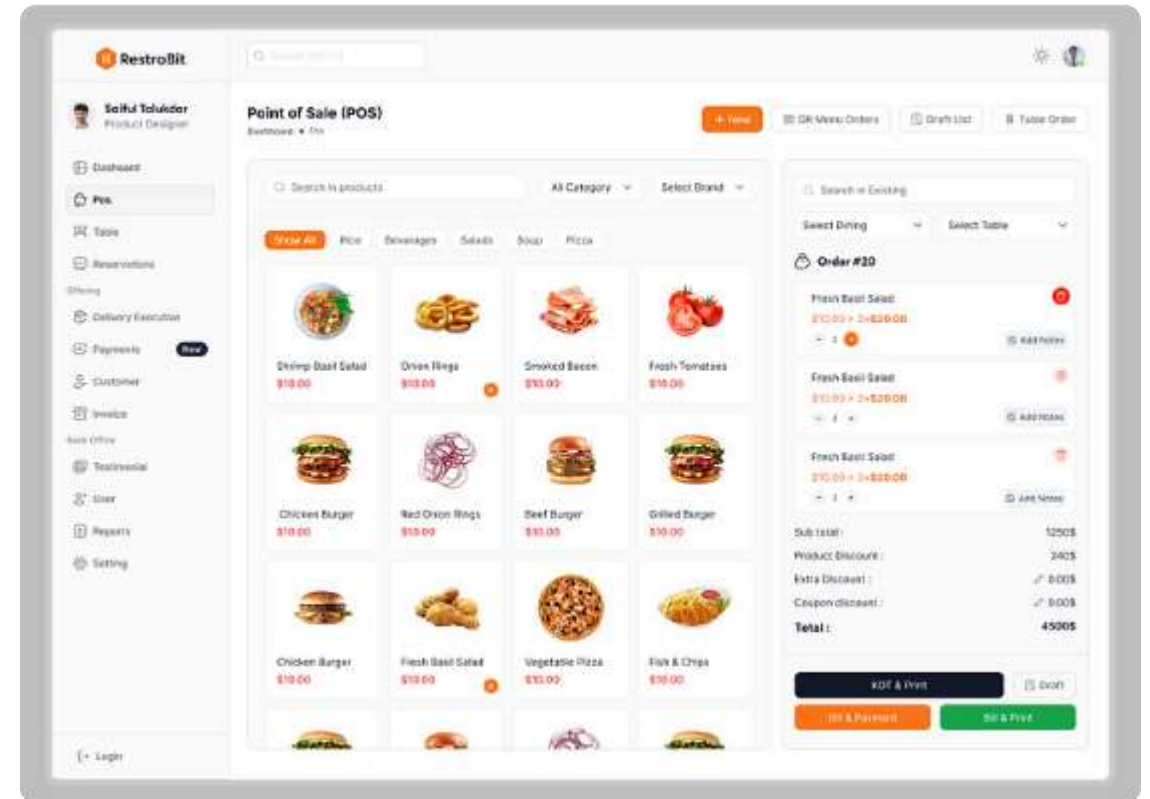
Cloud Bites is a full-stack web application designed for high availability and performance.

Customer Side

Register, Search Food, Cart Management, Order Tracking.

Admin Side

Menu Control, Dashboard Analytics, Order Status Updates.



Technology Stack

Component	Technology Used	Brief Description
Frontend	HTML, CSS, Bootstrap, JS	Responsive UI with modern, clean layouts.
Backend	PHP	Server-side scripting for logic and DB connection.
Server	Apache	High-performance web server hosted on EC2.
Database	MySQL	Relational data storage managed via Amazon RDS.
Cloud Platform	Amazon Web Services (AWS)	Scalable cloud infrastructure & networking.

Core AWS Infrastructure

Amazon EC2

Hosts the Apache Web Server and PHP application files. Provides reliable compute capacity.

Amazon RDS

Managed MySQL database storing users, menu, and orders. Offloads DB administration.

Amazon VPC

Logically isolated virtual network. Houses EC2 and RDS for secure communication.

Security Groups

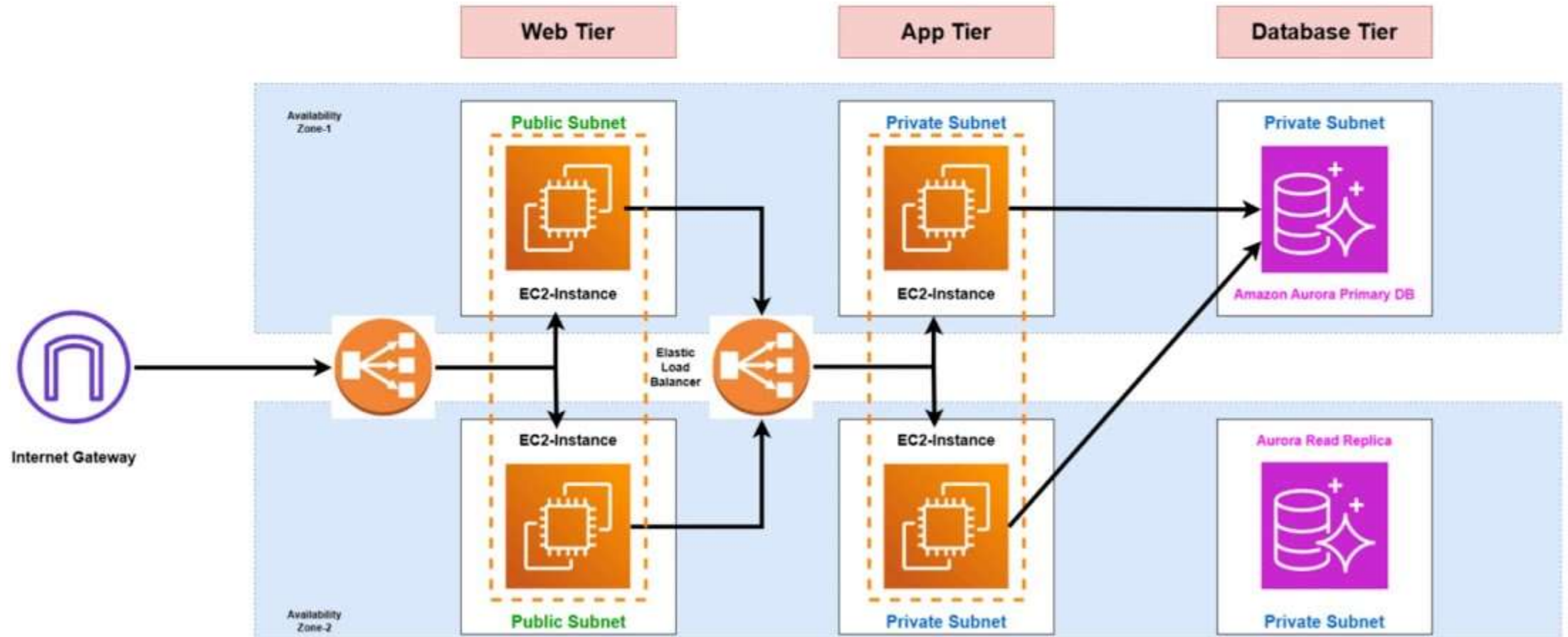
Virtual firewalls controlling traffic. Allowed Ports: 22 (SSH), 80 (HTTP), 3306 (MySQL).

Internet Gateway

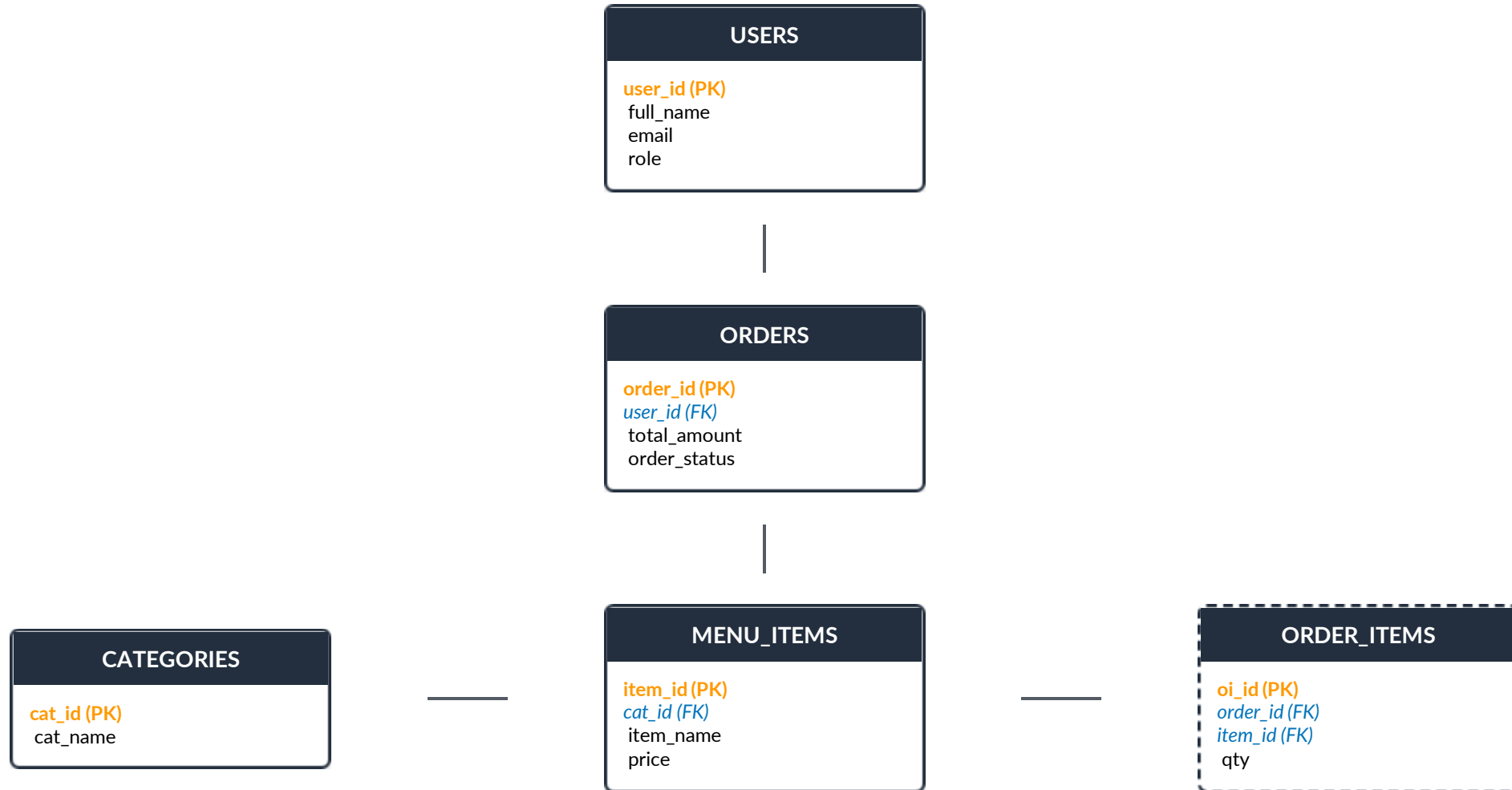
Bridge between VPC and the public internet, enabling users to access the application.

Public IP

Unique identifier used to access the application directly via the browser.



Database Design & Relationships



Note: *ORDER_ITEMS* acts as a junction table connecting multiple items to a single order.

Project Implementation Steps



Deployment Flow: Instance Setup → Environment Provisioning → Database Linking → Application Upload → System Integration Testing.

System Features

Customer Module

- Secure Registration & Login
- Search Bar & Category Browsing
- Dynamic Add to Cart
- Order History Tracking
- Responsive Web Interface

Admin Module

- Centralized Dashboard
- Manage Menu & Upload Food Images
- Real-time Order Management
- Update Order Status (Preparing/Sent)
- Customer Record Management

Future Scope (Roadmap)

AWS Route 53

Implement custom domain mapping (e.g., cloudbites.com).

AWS ACM & S3

SSL Certificates for HTTPS and S3 for static image storage.

ELB & Auto Scaling

Distribute traffic across instances and scale based on load.

CloudWatch & SNS

Resource monitoring and SMS/Email order alerts.

Payment Gateway

Integration with UPI, Stripe, or PayPal for live transactions.

Mobile App

Native Android and iOS applications using AWS Amplify.

Results

100%
AWS Deployed

Key Achievements

- Successful integration of EC2 and RDS
- Role-based Access Control (Admin/Customer)
- Efficient dynamic menu search logic
- Secure environment via Security Groups

Conclusion

Cloud Bites successfully demonstrates how AWS services like EC2, RDS, and VPC work together to create a secure, scalable, and efficient restaurant infrastructure.

Thank You

Questions?